Akoizo Original Practice - Form 1 - Answer key

Division C | Circuit Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) A negative charge's force is…

Unit: Electrostatics, historical ideas, and event reasoning.
Worked-case data: Two point charges keep their values while separation triples. Predict force.

A. Opposite the electric field

B. Always with the field

C. Always zero

D. Independent of its charge

Correct: A

2. (2 points) An ampere is…

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

A. Joule per coulomb

B. Ohm per meter

C. Coulomb per second

D. Joule per second

Correct: C

3. (2 points) Parallel resistors share…

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

A. Voltage

B. Always current

C. Always power

D. Always resistance

Correct: A

4. (2 points) An LED’s forward voltage is…

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

A. Always zero

B. Equal to any source

C. Device- and condition-dependent

D. Always 0.7 V

Correct: C

5. (2 points) XOR(1,1) equals…

A. 0

B. 1

C. 2

D. Undefined

Correct: A

6. (2 points) KCL expresses…

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

A. Energy creation

B. Resistance addition only

C. Charge conservation

D. Equal current in all branches

Correct: C

7. (2 points) A voltmeter connects…

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

A. Across two nodes

B. As an intentional source short

C. Only instead of every wire

D. Only to ground alone

Correct: A

8. (2 points) Equal LED current guarantees equal brightness?

Unit: LED matching, charge-time estimates, and troubleshooting.
Worked-case data: Estimate ideal time for 1.5 Ah supplied at constant 0.30 A.

A. Only for any red LED

B. Only at 5 V

C. No

D. Always

Correct: C

9. (2 points) A D flip-flop mainly adds…

Unit: Transistors, op-amps, storage, and timing.
Worked-case data: An ideal noninverting amplifier has Rf = 9 kΩ and Rg = 1 kΩ. Predict gain.

A. Stored state

B. Unlimited voltage

C. Zero timing constraints

D. Only resistance

Correct: A

10. (2 points) Tripling distance changes force to…

Unit: Electrostatics, historical ideas, and event reasoning.
Worked-case data: Two point charges keep their values while separation triples. Predict force.

A. One third

B. Nine times

C. One ninth

D. Three times

Correct: C

11. (2 points) A watt-hour measures…

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

A. Energy

B. Current

C. Resistance

D. Charge alone

Correct: A

12. (2 points) Two 100 Ω resistors in series give…

A. 100 Ω

B. 10000 Ω

C. 200 Ω

D. 50 Ω

Correct: C

13. (2 points) A series resistor mainly…

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

A. Limits current

B. Creates charge

C. Guarantees any voltage is safe

D. Reverses polarity

Correct: A

14. (2 points) NOT(A AND B) equals…

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

A. (NOT A) AND B

B. A AND B

C. (NOT A) OR (NOT B)

D. A OR B

Correct: C

15. (2 points) Negative solved current means…

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

A. Opposite the assumed direction

B. Impossible circuit necessarily

C. Negative charge disappears

D. No voltage exists

Correct: A

16. (2 points) An ohmmeter generally requires…

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

A. Maximum possible voltage

B. No instrument instructions

C. A de-energized suitable circuit

D. A live mains circuit

Correct: C

17. (2 points) 1.5 Ah / 0.30 A equals…

A. 5 h

B. 0.45 h

C. 5 J

D. 0.20 h

Correct: A

18. (2 points) The op-amp virtual short requires…

Unit: Transistors, op-amps, storage, and timing.
Worked-case data: An ideal noninverting amplifier has Rf = 9 kΩ and Rg = 1 kΩ. Predict gain.

A. Saturation

B. Disconnected inputs

C. Suitable negative feedback and linear operation

D. Any circuit whatsoever

Correct: C